An RRAM Digital Computing-in-Memory Macro With Dual-Mode Multiplication and Maximum Value Rounding Adder Tree

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***Abstract*—Implementing digital computing-in-memory (DCIM) based on resistive memory (RRAM) faces several critical challenges due to the small signal margin, large device variations, and large energy- and area-overhead induced by the digital adder tree (AT). To address these issues, we propose an RRAM DCIM macro based on the standard foundry one-transistor-one-resistor (1T1R) cell array featuring: 1) dual-mode MAC operation for efficiency- or accuracy-oriented optimization; 2) margin-enhanced digitized unit (MEDU) to amplify the signal ratio; and 3) maximum value rounding AT (MVR-AT) to reduce its power- and area-overhead. A test chip is demonstrated using a 180 nm CMOS process to verify the concept. It achieves a peak energy efficiency (EF) of 63.08 TOPS/W in the efficiency-oriented mode and a minimum error rate of 1.58% in the accuracy-oriented mode. Their combination can meet the requirements of different workloads in AI computing tasks to optimize the overall power consumption with negligible accuracy loss.**

***Index Terms*—Approximate adder tree (AT), digital computing-in-memory (DCIM), dual-mode operation, margin enhancement, resistive memory (RRAM).**

I. Introduction

With the skyrocketing development of intelligent computing, the memory wall issue has been brought to the front line and become the major performance bottleneck [1]. New paradigms and technologies that can minimize data movement are being intensively studied [2], [3]. Among the emerging solutions, digital computing-in-memory (DCIM) [4], [5], [6], [7], [8], [9], [10], [11], which integrates logic circuitry inside the memory array, is under intensive studied for AI model processing. It has been demonstrated on cutting-edge technology nodes [8] as well as integrated into advanced system-on-chip (SoC) processors [9]. Compared to its analog counterpart, DCIM has exhibited advantages in several aspects. First, its accuracy is much less subjective to process variations as it performs computing in the digital domain. Second, it can achieve high energy efficiency (EF) by eliminating the power used for the conversion between analog and digital signals. Finally, its performance density can be effectively boosted by leveraging the relentless progress of advanced technology nodes. Most of the recent DCIM chips [4], [5], [6], [7], [8], [9] are implemented using SRAM, which can be seamlessly integrated with logic gates and adder trees (ATs). Despite the above advantages, SRAM DCIM tends to have limited capability compared to that based on the nonvolatile memory (NVM), mostly because the SRAM cell has relatively low bit-cell density [4], [5], [6] and non-negligible stand-by power [12]. Besides, although the conventional NVM is not available below the 28/22-nm process, emerging NVM (eNVM), such as resistive memory (RRAM), is under development for advanced technology nodes [13]. Consequently, eNVM can potentially become a competent solution to improve the on-chip data accommodating capability of DCIM-based systems.

Implementing DCIM on eNVM is a non-trivial issue considering several aspects. First of all, unlike the active SRAM cells, the passive eNVM cells cannot directly output the rail-to-rail signals (ground or VDD). In previous arts, the output signals are generated by discharging the parasitic nodes depending upon RRAM cell resistance [10], [11]. However, the signal margin is relatively small because of the limited resistance-ratio (R-ratio) of RRAM cells which is defined as the ratio of the high to low resistance states of the RRAM. Besides, eNVM cells usually suffer from large device variations [14], which further degrades the signal margins and induces considerable computing errors. There are eNVM array architectures [15], [16] based on dual eNVM cells for highly accurate as well as energy-efficient read operations. Moreover, to maintain the high-density advantages of eNVM, it is of particular importance to reduce the hardware cost of the logic circuitry integrated into the cell array. To deal with this issue, recent advances have proposed using approximated ATs, such as replacing the first stage of the AT with logic OR gates [11] or introducing customized 4-2 compressors [17], yet using radical approximation strategies may incur considerable accuracy loss. Because of these challenges, eNVM-based DCIM designs have rarely been demonstrated in silicon.

In this work, we proposed an RRAM DCIM design based on the standard one-transistor-one-resistor (1T1R) cell array featuring dual-mode multiply-and-accumulate (MAC), margin-enhanced digitized unit (MEDU), and maximum value rounding AT (MVR-AT). The details of the proposed schemes are discussed in Section II. The performance evaluation and silicon-verified results are shown in Section III. Section IV concludes this work.

II. Proposed RRAM-Based DCIM Macro

Fig. 1 shows the overall structure of the proposed macro, which consists of the 1T1R cell array integrated with MEDU and MVR-ATs, drivers for word-line (WL), bitline (BL), source-line (SL), and input, timing, and mode controller (Ctrl). The macro is divided into multiple sub-arrays to improve computing throughput. Each sub-array is further divided into 128 groups. Each group has 32 × 4 1T1R cells, in which 4-bit weight can be mapped into different columns. Depending on the states of the transmission gate, the local SL (LSL) can be connected either to the global SL (GSL) for pre-charge, read, and write or to the MEDU to multiply with the input (IN) data. The MVR-AT accumulates the 4-bit multiplication results from 128 groups ($D_0[0:3]$~$D_{127}[0:3]$) to generate the 11-bit output in a single clock (CLK) cycle.

A. Proposed Dual-Mode MAC Operations With MEDU

The macro can perform MAC operation either in the 1T1R mode for high EF or in the paired-1T1R (P-1T1R) mode for low error rate. The details of MEDU and the 1T1R MAC mode are shown in Fig. 2(a). It consists of one pMOS (PM1) to pre-charge the output line (OUTL), two NMOSs (NM1 and NM2) to discharge OUTL depending on the input and weight signals applied on their gates, and one data flip-flop (DFF) to latch the output data (DOUT). Notice that the NM1 is used to sense the voltage on LSL (V_X)

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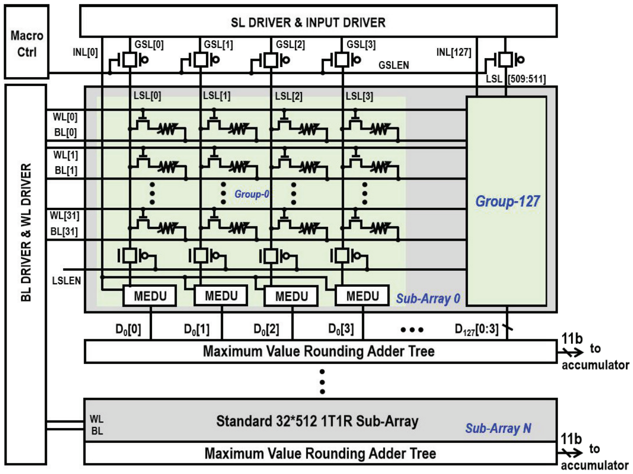


Fig. 1. Overall structure of the proposed 1T1R-based RRAM DCIM macro featuring dual-mode multiplication, MEDU and MVR-AT.

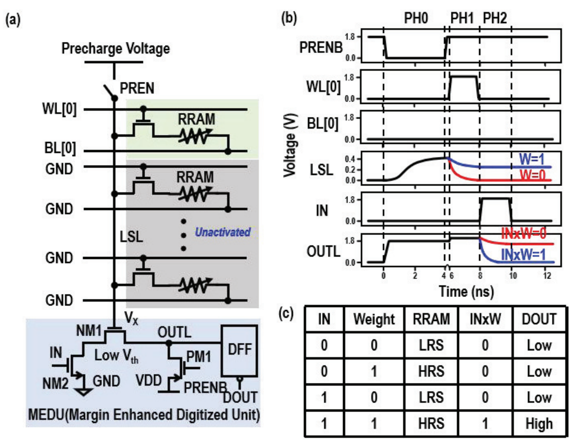


Fig. 2. Illustration of 1T1R MAC mode. (a) Conceptual view. (b) Operational waveforms. (c) Input and weight data encoding scheme. LRS and HRS represent the high- (HRS) and LRS, respectively.

determined by the weight (W) data and perform transconductance amplification of the change of V_X (ΔV_X). A typical computing cycle can be divided into three phases (PH0-PH2), which is shown in Fig. 2(b). In PH0 (PRENB = 0), the parasitic capacitance of the LSL (87 fF according to post-simulation) and OUTL is pre-charged to 0.4 V and VDD, respectively. The LSL pre-charge voltage is selected according to simulation results to maximize the signal margin. In PH1, the selected WL (WL [0]) is activated and the LSL starts discharge to the grounded BL (BL [0]). It discharges to a higher voltage level in the case of $W = 1$ compared to that in the case of $W = 0$. In PH2, the input data are applied to the gate of NM2. Consequently, the OUTL voltage is developed to a low or high value, determined by the multiplication result of IN and W. Fig. 2(c) shows the input and weight data encoding scheme.

Fig. 3(a) shows the conceptual view of the P-1T1R mode. Its MAC operation is similar to that in the 1T1R mode excepting a pair of RRAM cells are involved. The weight is complementarily mapped in P-1T1R mode, as shown in Fig. 3(c). The operational waveforms of the P-1T1R MAC mode are shown in Fig. 3(b). During PH0, the OUTL is pre-charged to VDD, and the LSL is pre-charged to a lower voltage (0.3 V) compared to the 1T1R mode because of the additional charging phase (PH1) which will incur a negligible delay. In PH1, the WL [0] is activated to pull up the LSL through one of the activated BL (BL [0]), which is set to 0.6 V. Subsequently, in PH2, WL [1] is activated to pull down the LSL through the other BL (BL [1]), which is set to ground (GND). The voltage of the LSL is determined by the pull-up and the pull-down strength due to the weight data. The two voltage-developing phases (PH1 and PH2) in the P-1T1R mode are used here to enlarge the voltage swing of Vx. In PH3, the OUTL discharges according to the multiplication result of IN and W.

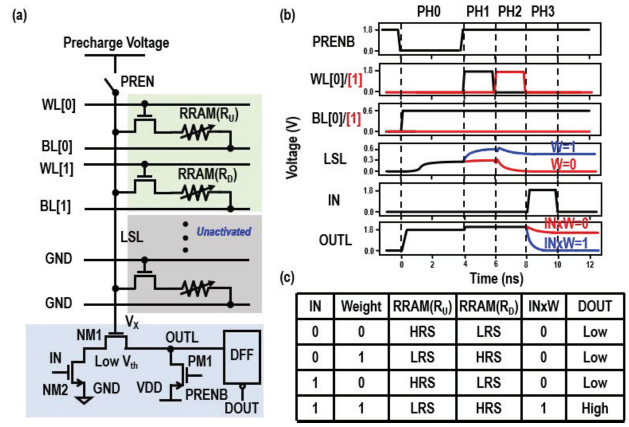


Fig. 3. Illustration of P-1T1R MAC mode. (a) Conceptual view. (b) Operational waveforms. (c) Input and weight data encoding scheme.

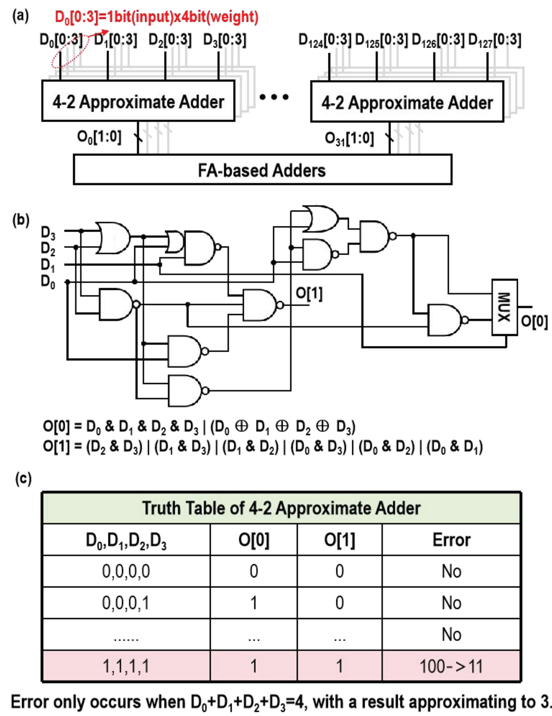


Fig. 4. Proposed MVR-AT. (a) Overall structure. (b) Schematic of 4-2 approximate adder. (c) Its truth table.

B. Proposed MVR-AT

Fig. 4(a) shows the structure of the proposed MVR-AT. It consists of 128 4-2 approximate adders used for the first stage summation. The 4-2 approximate adder achieves data width compression by neglecting the maximum summation value. Each 4-bit addend (D_0[0:3]) from the MEDUs in the same group is separately sent to four approximate adders. The 128 4-bit addends ($D_0 \sim D_{127}$) are bit-wisely sent to the approximate adders, which output 32 2-bit

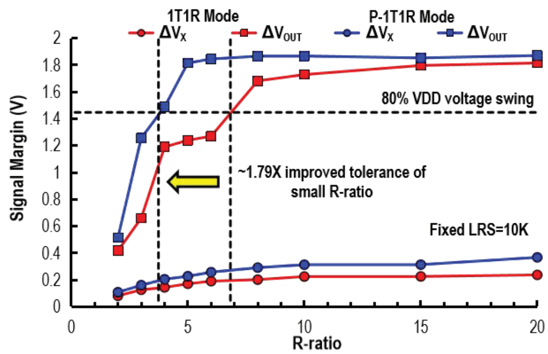


Fig. 5. Comparison on the signal margin as a function of R-ratio in the 1T1R and P-1T1R MAC modes.

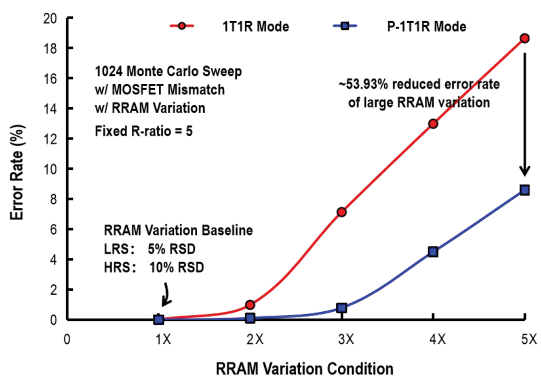


Fig. 6. Comparison on multiplying error rates in the 1T1R and P-1T1R MAC modes.

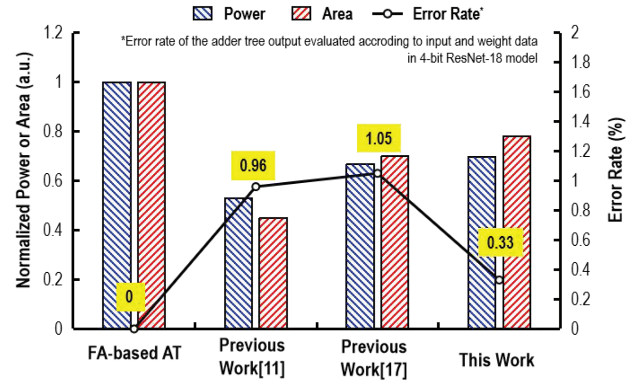


Fig. 7. Comparison on power, area and error rates of different ATs, including FA-based AT, previously approximate AT and the proposed MVR-AT.

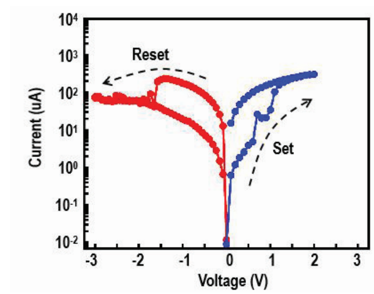


Fig. 8. Measured RRAM I–V curves of set and reset operation.

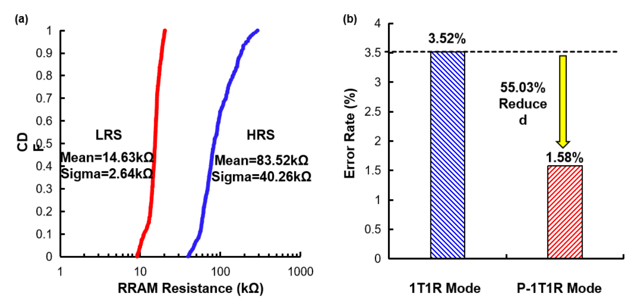


Fig. 9. (a) Measured RRAM resistance distributions. (b) Multiplying error rates of the RRAM DCIM macro in the 1T1R and P-1T1R modes.

results (O_0[0:1]~ O_{31}[0:1]), These 2-bit outputs are then sent to FA-based adders for further accumulation.

Fig. 4(b) shows the schematic of the 4-2 approximate adder and its logic expressions. Four 1-bit addends (D_0, D_1, D_2, D_3) are added up at the 4-2 approximate adder to output a 2-bit approximate result (O[0:1]). The corresponding truth table of the proposed 4-2 approximate adder is shown in Fig. 4(c). It is worth mentioning that the errors only occur when all four addends are one. As a result, the maximum value (100) of the approximate adder will be approximated to a lower value (11).

III. Results and Discussion

A. Comparison and Analysis

Fig. 5 shows the comparison of the signal margin as a function of R-ratio in the 1T1R and P-1T1R MAC modes based on fixed low-resistive-state (LRS). Comparing the voltage swings of OUTL (ΔV_{OUT}) and ΔV_X, it can be seen that ΔV_X is effectively amplified by the MEDU. Besides, thanks to the paired charging and discharging phases, the P-1T1R mode exhibits a larger signal margin across all R-ratios compared to the 1T1R mode. At an R-ratio of 7 or above, ΔV_{OUT} in both the 1T1R and P-1T1R modes exceeds 80% of the VDD voltage swing. At smaller R-ratios, the P-1T1R mode achieves 1.79 times the improved tolerance compared to the 1T1R mode.

Fig. 6 shows the comparison of multiplying error rates in the 1T1R and P-1T1R MAC modes considering both the MOSFET mismatch and RRAM variation. The R-ratio is fixed at 5 in this evaluation considering the current foundry RRAM usually suffers from a small R-ratio [18] as well as the trade-off between the R-ratio and the endurance [19]. Unlike the 1T1R scheme, the multiplication results in the P-1T1R mode are determined by the differential resistance of paired RRAM cells, making it less susceptible to process variation and can effectively mitigate the impact of RRAM variations on the signal margin [20]. The P-1T1R mode has a lower multiplying error rate compared to 1T1R mode across all RRAM variation conditions. Considering the RRAM cells having 25% and 50% relative standard deviation (RSD) in LRS and HRS, the P-1T1R mode achieves a 53.93% reduced error rate compared to the 1T1R mode.

Fig. 7 shows the comparison between the proposed MVR-AT and the conventional FA-based AT, as well as previous approximate ATs [11], [17]. The power and area of the MVR-AT are reduced by 30.29% and 22.08%, respectively, compared to the FA-based AT. The proposed MVR-AT also shows a lower summation error rate (0.33%) compared to other approximate ATs, as evaluated in a typical inference task using a 4-bit ResNet-18 model. We speculate that this is due to the prevalent IN and weight sparsity in the neural network model.

B. Measurement Results

The key components of the proposed macro are verified using a 180 nm CMOS process with back-end-of-line (BEOL) integrated

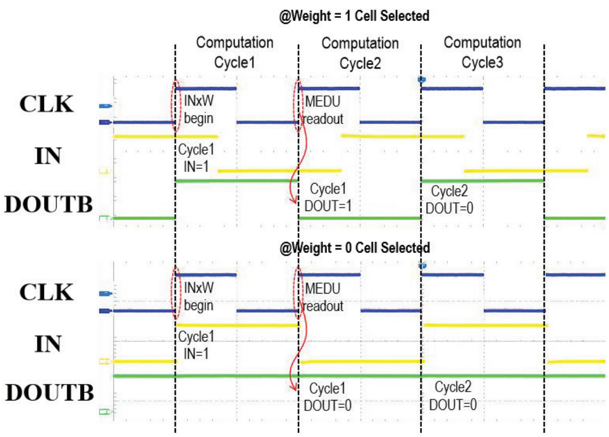


Fig. 10. Measured typical functional waveforms of the RRAM DCIM macro.

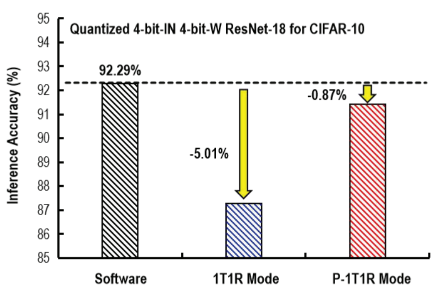


Fig. 11. Simulated inference accuracy of the proposed RRAM DCIM macro based on a modified ResNet-18 model trained on the CIFAR-10 dataset.

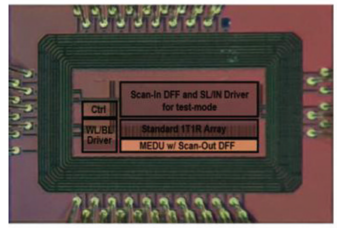
oxide RRAM. It consists of a 32 × 512 standard 1T1R array, 512 MEDUs, drivers and addressing circuits, DFF chain for data input and output. Fig. 8 shows the typical I-V switching curve of the RRAM cell. Fig. 9(a) shows the measured RRAM resistance distribution. A 55.05% reduction in the multiplying error rate is achieved after applying the P-1T1R mode compared to the 1T1R mode, as shown in Fig. 9(b).

Fig. 10 shows the typical captured functional waveform of the RRAM DCIM macro at different weight addresses with DOUT in both 0 and 1 conditions. The multiplying result of a given CLK cycle is output at the beginning of the next cycle. During testing, the clock cycle time is 100 μs, which is used to validate our scheme. This design allows the macro to work with increased CLK frequency by pipelining the multiplication and accumulation operation.

To evaluate the impact of the dual-mode MAC operation and the MVR-AT on typical DNN tasks, we performed simulations on the inference accuracy using an open-source benchmark framework for CIM accelerators [21] with a modified quantized method, input/weight/output precisions and multiplying error rate of dual-mode operation. Fig. 11 compares the inference accuracy in 1T1R and P-1T1R mode. The inference accuracy drop is 5.01% in 1T1R mode and 0.87% in P-1T1R mode. It can be seen that the P-1T1R mode effectively reduces the inference error rate.

Fig. 12 shows the photograph and summary table of the proposed RRAM DCIM macro. The measured energy per multiply is 10.11fJ/multiply in 1T1R mode and 15.16fJ/multiply in P-1T1R mode.

Table I compares the proposed RRAM DCIM macro with recent RRAM-based macros. Compared to previous works, our design features MEDU for margin enhancement and the P-1T1R mode for improved variation tolerance. In addition, the MVR-AT is introduced to reduce energy and area costs with a low error rate. Finally, based on the post-simulation, the normalized EF is competitive compared to previous works.



Chip Summary	
Technology	180nm UMC Generic II Process
RRAM	TaOx-based RRAM
Bit-Cell	1T1R /P-1T1R
Macro Size	2000 × 1400μm^2
Array Size	32 rows × 512 columns (16Kb)
Power Supply (VDD)	1.8V (Digital), 3.3V (Analog)
Energy Efficiency	10.11fJ/multiply (1T1R) 15.16fJ/multiply (P-1T1R)

Fig. 12. Die photograph and chip summary.

TABLE I

COMPARISON TABLE

	JSSC 2022[14]	TCASI 2022[10]	TCASII 2023[11]	This Work
Technology	40nm	65nm	28nm	180 nm
Memory	RRAM	RRAM	RRAM	RRAM
Bit-Cell	1T1R	3.25T1R	1T1R	1T1R/P-1T1R
Bitcell Area	N.A.	0.847um^2	0.229um^2	2.052um^2
Array Dimensions	64kb	64kb	16kb	16kb
CIM Domain	Analog CIM	Digital CIM	Digital CIM	Digital CIM
Silicon Verified	Yes	Post Simulation	Post Simulation	Yes
Margin Enhancement	N.A.	N.A.	N.A.	MEDU (1.79x enhanced margin)
Variation Tolerance	Write Verification	N.A.	N.A.	Paired-1T1R mode (53.93% error rate reduction)
Adder Tree Scheme	N.A.	FA-based AT	OR gate approximate AT	Maximum Value Rounding AT
Normalized Energy Efficiency (TOPS/W)	56.67	114.41	1966	63.08(1T1R mode)* 54.39(P-1T1R mode)*

*Energy efficiencies based on post-simulation

IV. CONCLUSION

In this work, we propose a silicon-verified 180 nm RRAM DCIM macro based on the standard foundry 1T1R cell array. We introduce the dual-mode MAC operations, which include the 1T1R mode to improve the EF and the P-1T1R mode to reduce the error rates. In addition, we propose the MEDU amplify the signal margin, which can directly output rail-to-rail signals with a small area overhead. Furthermore, we proposed the MVR-AT that performs accurate summation excepting the maximum value. It can reduce energy and area by 30.29% and 22.08%, respectively, with only a 0.33% error rate in typical neural network inference tasks. The key components of the proposed macro are verified using a 180 nm CMOS process with BEOL-integrated oxide RRAM. The test chip can achieve a peak EF of 63.08 TOPS/W in 1T1R mode and 54.39 TOPS/W in P-1T1R mode. The multiplying error rate is 3.52% in 1T1R mode and 1.58% in P-1T1R mode. The proposed concepts and schemes discussed in this work can be easily extended to the DCIM design based on other types of emerging memory, which can potentially result in nonvolatile DCIM-based AI accelerators.

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